CSE STUDY HUB

DATA COMMUNICATION & NETWORKING UNIT 4

Transport Layer

SOURCE-ALIGNED CONCISE NOTES

Prepared using the uploaded handwritten class notes for this unit. The explanations retain the concepts and examples visible in those notes, with wording cleaned up for revision.

Topics covered in this unit

- Process-to-process delivery and ports
- UDP and TCP
- TCP sequence numbers, ACKs and checksum
- Connection setup, flow control and congestion

How to use: Learn the definition, draw the diagram/flow where applicable, practise the numerical method, and finish with the revision questions. If a detail is not present in the uploaded pages, it is not added as an asserted class-note topic.

Exam answer structure

For a 5-mark answer: definition → labelled diagram/formula → working or components → example/application → advantage, limitation or comparison.

UNIT 4 · Process-to-process delivery and ports

The Transport Layer provides process-to-process communication. The Network Layer delivers packets to a host; the Transport Layer uses port numbers to deliver data to the correct application process.

A port number is 16 bits. A socket endpoint combines an IP address and a port number; a communication flow is identified using source and destination IP addresses and port numbers, along with the transport protocol.

Well-known service ports in the notes include HTTP 80, HTTPS 443, DNS 53, SMTP 25 and FTP 20/21. Port numbers identify services/process endpoints, not the physical computer itself.

Transport data units are commonly called segments for TCP and datagrams for UDP. The transport header is added around application data.

UNIT 4 · UDP and TCP

UDP is connectionless and has a small header. It does not establish a connection or guarantee delivery, ordering or retransmission. It is useful when low overhead and timely delivery matter, such as real-time media or simple request/response traffic.

TCP is connection-oriented and provides reliable, ordered byte-stream delivery. It uses sequence numbers, acknowledgements, checksums, retransmission and flow control.

TCP and UDP both use port numbers for process identification. TCP adds connection management and reliability mechanisms; UDP keeps the transport service simpler and lighter.

Choose TCP when reliable ordered delivery is required. Choose UDP when the application can tolerate loss or handle it itself and low delay/overhead is important.

UNIT 4 · TCP sequence numbers, ACKs and checksum

TCP sequence numbers identify byte positions in the stream. The acknowledgement number indicates the next byte expected from the other side.

An ACK confirms receipt of data up to the acknowledged point. If data is lost, TCP can retransmit it. Sequence numbers also help detect duplicates and put received bytes back in order.

The TCP checksum is used for error detection over the TCP segment and a pseudo-header. It helps detect corruption; it is not encryption and does not provide confidentiality.

Sequence and acknowledgement numbers are measured in bytes, not simply as a count of segments. This is important when interpreting TCP examples.

UNIT 4 · Connection setup, flow control and congestion

TCP connection establishment uses a three-way handshake: (1) client sends SYN, (2) server replies SYN-ACK, (3) client sends ACK. This synchronizes initial sequence numbers and establishes the connection.

TCP flow control uses the receiver-advertised window to prevent the sender from overflowing the receiver's buffer. The sender limits the amount of unacknowledged data in flight.

Congestion control protects the network from excessive traffic. TCP adjusts its sending rate using mechanisms such as slow start and congestion avoidance, responding to signs of congestion such as loss.

Flow control protects the receiver; congestion control protects the network. This distinction is a common exam comparison.